

1 Planning (P)

Chlorine gas, when bubbled into hot dilute aqueous potassium hydroxide, produces a solution containing potassium chlorate(V) and potassium chloride.

- (a) Write a balanced equation for the reaction between chlorine and potassium hydroxide.

..... [1]

Evaporation of the resulting solution gives a white solid which is a mixture of potassium chlorate(V) and potassium chloride.

Pure potassium chloride can be separated from this mixture by a process known as **fractional crystallisation**. In this process, the substances are separated based on their difference in solubility.

The **solubility** of a solid is defined as:
 'The mass of solid that will dissolve in and just saturate 100 g of solvent (water).'

- (b) Most solids become more soluble as temperature increases.

Table 1 below shows the solubility of potassium chlorate(V) and potassium chloride over a range of temperature.

temperature / °C	10	20	30	40	60	80	100
solubility of potassium chlorate(V), KClO ₃ , / g per 100 g of H ₂ O	5.0	7.0	10.5	14.0	24.5	38.5	57.0
solubility of potassium chloride, KCl, / g per 100 g of H ₂ O	31.0	34.0	37.0	40.0	45.5	51.1	56.7

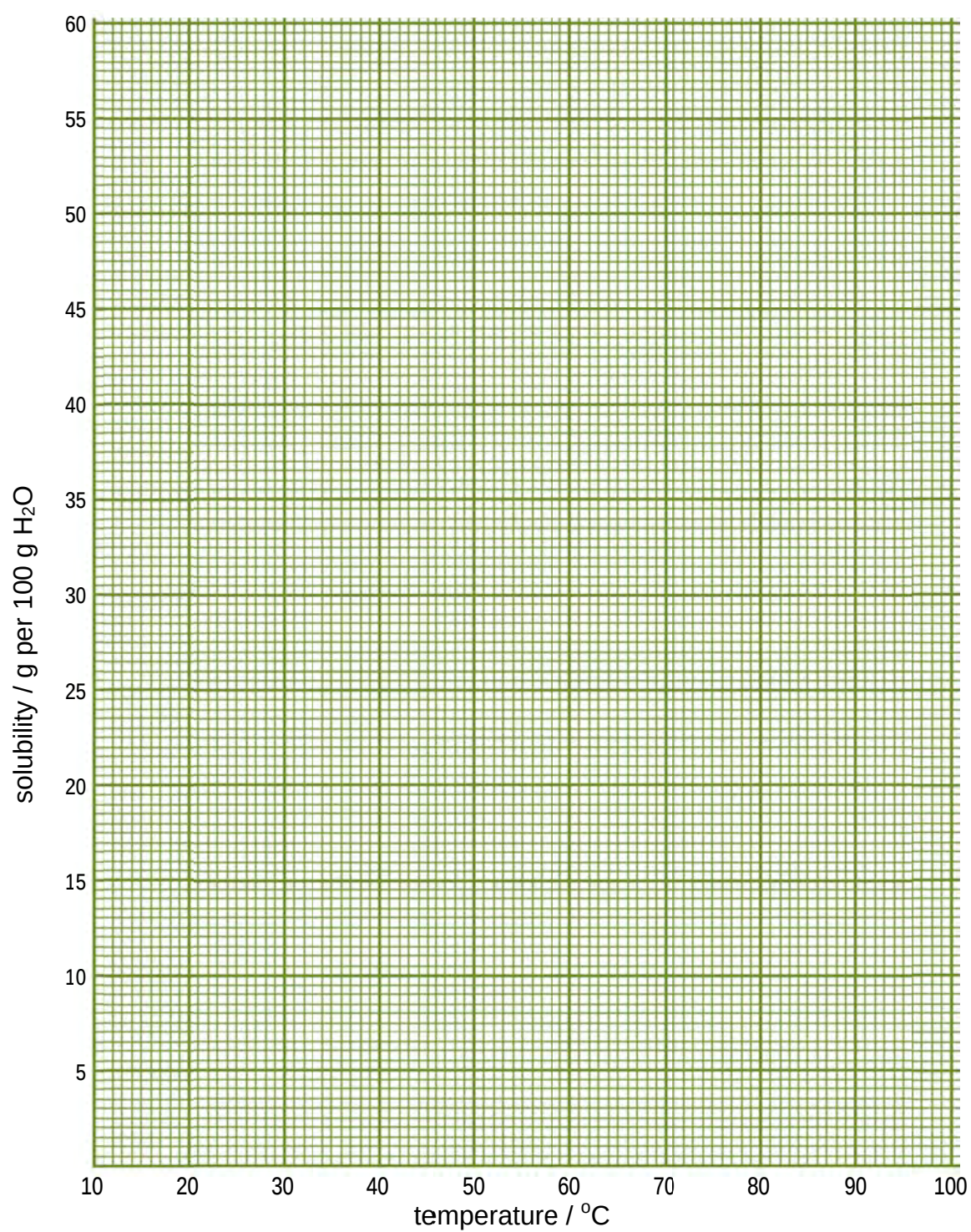
Table 1

Plot the data in **Table 1** on the grid on the next page and draw the solubility curves for each substance.

Label the solubility curves clearly.

[3]

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(c) In the process of **fractional crystallisation**, the solid mixture is first dissolved in heated solvent to obtain a solution. The resultant solution is then carefully cooled to a selected temperature to enable one solid to crystallise while ensuring the other solid remains dissolved in the solution.

You may assume that all of the solid should be dissolved in 100 g of water at the start of the separation.

- (i) From the graph in (b), obtain the minimum temperature to which the solution must be heated in order to dissolve all of the 16.4 g of potassium chlorate(V) and 50.0 g of potassium chloride.

(ii) Write a plan to obtain a maximum amount of potassium chloride, KCl , free from potassium chlorate(V), $KClO_3$, from the mixture that contains **16.4 g** of potassium chlorate(V) and **50.0 g** of potassium chloride.

Your plan should include:

- the essential details of the general fractional crystallisation procedure, including the capacity of apparatus used (where relevant)
- the temperature to which the solution is to be cooled and the justifications for your choice (reference to the graph in **(b)** is required)
- a precautionary step you would take to improve the purity of the potassium chloride sample crystals that are obtained

[illegible]

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- (d) With reference to the graph in (b), calculate the mass of potassium chloride, KCl, you would expect to recover.

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- (e) How could you check the purity of the sample of potassium chloride that has been obtained?

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[1]

[Total: 12]

- 2 The following table lists the standard enthalpy changes of combustion, $\Delta H_c^\ominus$, of some monohydric alcohols.

alcohol	$\Delta H_c^\ominus / \text{kJ mol}^{-1}$
methanol	-715
ethanol	-1367
propan-1-ol	
butan-1-ol	-2671

[Turn over

- (a) Write an equation to represent the standard enthalpy change of combustion of propan-1-ol.

..... [1]

- (b) (i) The difference in the standard enthalpy change of combustion of methanol and ethanol is -652 kJ mol^{-1} . By considering the structures of the two alcohols, suggest the significance of this difference.

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- (ii) Hence suggest a value of $\Delta H_c^\ominus$ for propan-1-ol.

$\Delta H_c^\ominus$ for propan-1-ol = kJ mol^{-1}

- (iii) Given $\Delta H_f^\ominus (\text{H}_2\text{O}(l)) = -286 \text{ kJ mol}^{-1}$ and $\Delta H_f^\ominus (\text{CO}_2(g)) = -394 \text{ kJ mol}^{-1}$ calculate the standard enthalpy change of formation of propan-1-ol, $\Delta H_f^\ominus (\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}(l))$.

[4]

- (c) Draw a labelled diagram to show the interactions between ethanol molecules.

[2]

[Total: 7]

- 3 (a) A student electrolyses a solution of brine (concentrated aqueous sodium chloride) using platinum electrodes. She ensures that the solution is continually stirred at a temperature of not more than 40°C.

It was found that as electrolysis proceeds, sodium chlorate(I), NaClO, is formed in the electrolyte.

- (i) Suggest equations for the reactions occurring at the anode and cathode.

anode:

cathode:

- (ii) Write a suitable equation which shows how the NaClO could have been formed.

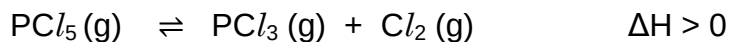
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- (b)** Sodium chlorate(I), NaClO , which is found in household cleaning products, is usually manufactured from chlorine gas.

When gaseous phosphorous (V) chloride is heated to 473 K in a closed container, the following equilibrium involving chlorine gas is established:



By using Le Chatelier's principle, predict and explain the effect of an increase in temperature on the above equilibrium.

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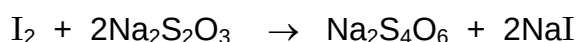
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[2]

- (c)** When sodium thiosulfate is reacted separately with chlorine and iodine, different products are formed.



- (i)** With reference to the change in oxidation state of sulfur in the reactions, deduce the relative oxidising strengths of chlorine and iodine.

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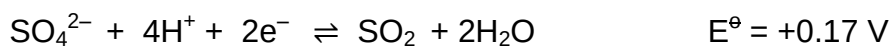
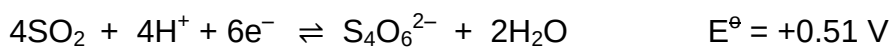
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- (ii) Use the following data, and data from the *Data Booklet*, to suggest the sulfur containing species that will be formed when bromine reacts with sodium thiosulfate.

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[Total: 9]

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- 4 Chromium is a typical transition element. Many of its compounds are intensely coloured.

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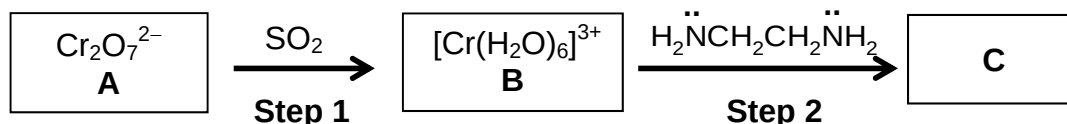
- (a) Write the electronic configuration for the chromium atom and chromium (III) ion.

Cr :

Cr³⁺ :

[2]

- (b) A reaction scheme involving chromium compounds is shown below:



The reaction in **Step 2**, which occurs spontaneously, involves 3 moles of the ligand reacting with 1 mole of **B** to form **C**.

- (i) Explain why chromium compounds are coloured.

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- (ii) What type of reaction occurs in **Step 1**?

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- (iii) Name the type of bond formed between the metal ion and the ligand in **Step 2**.

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- (iv) Write an equation for the reaction in **Step 2**.

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- (v) By considering the spontaneity of the reaction, together with the equation in (b)(iv), deduce the signs of ΔS , ΔH and ΔG for the reaction in **Step 2**.

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[9]

- (c) Silver chromate(VI) is a chromium containing compound which has applications in the field of photography.

The solubility product of Ag_2CrO_4 at 25°C is $1.12 \times 10^{-12} \text{ mol}^3 \text{ dm}^{-9}$.

- (i) Write an expression for the solubility product, K_{sp} , of Ag_2CrO_4 .

K_{sp} of $\text{Ag}_2\text{CrO}_4 = \dots\dots\dots$

- (ii) Calculate the solubility, in mol dm^{-3} , of Ag_2CrO_4 , at 25°C .

- (iii) Concentrated aqueous silver nitrate is added dropwise to $0.010 \text{ mol dm}^{-3}$ potassium chromate(VI).

What is the concentration, in mol dm^{-3} of silver ions when the first trace of precipitate appears?

[3]

[Total: 14]

[Turn over

- 5 Compound **X** is a disubstituted aromatic compound with the molecular formula $C_8H_9O_2N$ and contains 2 functional groups.

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Further investigations about compound **X** were conducted and the data about the reactions of **X** is shown in the table.

reaction	reagent	result
1	NaOH at room temperature	colourless solution formed
2	HCl at room temperature	no reaction
3	NaOH under reflux followed by careful acidification	two products were formed; one product had M_r of 152 and while the other was a gas that turned red litmus blue
4	Na	colourless gas evolved; white solid formed which is soluble in polar solvents
5	Br_2 (aq) in excess	Br_2 is decolourised; white solid formed with $M_r = 308.8$

Based on the information in the table and in this question, answer the following.

- (a) (i) Name the functional group that both reaction **1** and **5** show to be present in **X**.

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- (ii) Name the functional group present in **X** based on the results obtained in **both** reactions **2** and **3**. Explain your answer using **either** reaction **2** or **3**.

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[3]

- (b) Deduce the molecular formula of the white solid formed in reaction **5**.

[1]

(c) Draw a possible structural formula for the organic product in reaction 3.

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[1]

(d) (i) Use the given information, together with your answers in parts (a) to (c), draw the structural formula of X.

(ii) Explain clearly why you have placed **each** of the two functional groups in their particular positions.

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[Total: 8]

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- 6 Hydrocarbons are economically important chemicals due to their role as the world's main source of fuel, and as starting materials for the synthesis of many useful organic compounds.

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- (a) The table below gives the boiling point of three organic compounds of similar molecular mass.

Organic Compound	Structural Formula	M_r	Boiling point/ °C
hexane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	86.0	68
2-methylpentane	$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$	86.0	60
pentan-2-ol	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$	88.0	120

Explain the difference in boiling point between

- (i) hexane and 2-methylpentane

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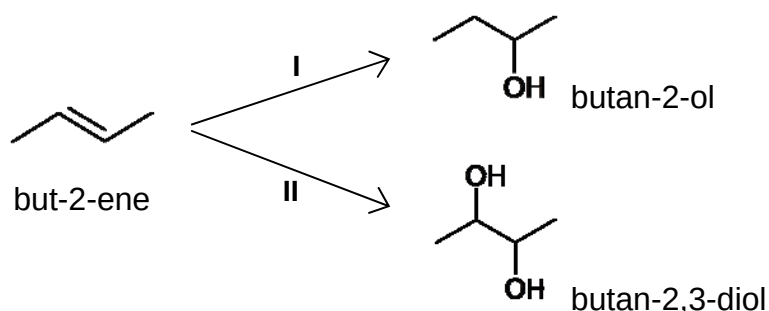
- (ii) 2-methylpentane and pentan-2-ol

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[4]

- (b) Alkenes of low molecular mass can be produced via the catalytic cracking of long chain hydrocarbons.

In the presence of suitable reagents, but-2-ene can react to produce alcohols such as butan-2-ol and butan-2,3-diol.



- (i) Suggest suitable reagents and conditions for the conversion of but-2-ene to

butan-2-ol

butan-2,3-diol

Butan-2-ol exhibits optical isomerism. However, the product obtained from step I does not rotate plane polarised light.

- (ii) Draw structural formulae to illustrate the optical isomers of butan-2-ol.

- (iii) Suggest why the butan-2-ol obtained in step I does not rotate plane polarised light.

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The effect of plane polarised light on butan-2,3-diol was investigated.

Three different types of butan-2,3-diol molecules were identified.

- Molecule **A** rotated plane polarised light to the left.
- Molecule **B** rotated plane polarised light to the right.
- Molecule **C** had no effect on plane polarised light.

- (iv) Suggest an explanation for the above observations made for molecules **A** and **C**.

Molecule	Explanation
A	
C	

[Turn over

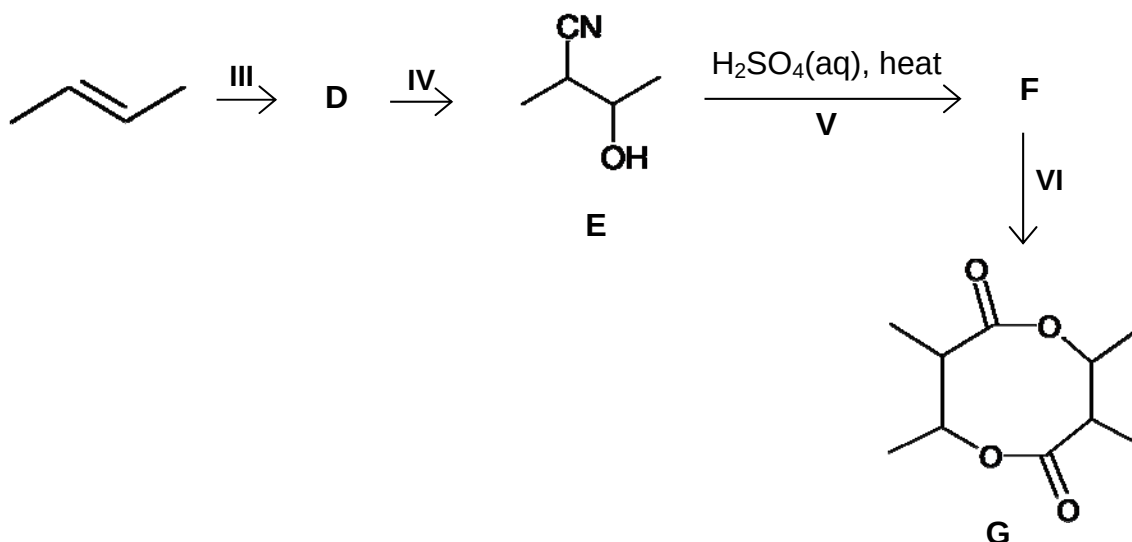
- (v) Suggest the ratio in which molecules of **A**, **B** and **C** are formed.

Ratio of **A**: **B**: **C**

[7]

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- (c) The following reaction scheme shows how but-2-ene can be used to synthesize a cyclic diester, **G**.



- (i) Suggest suitable reagents and conditions for steps **III**, **IV** and **VI**.

Step **III**

Step **IV**

Step **VI**

- (ii) State the type of reaction occurring in step **V**.

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- (iii) Suggest the structures of compounds **D** and **F**.

D	F

- (iv) Give the structural formula of the organic products formed when **E** is warmed with a solution of iodine and potassium hydroxide.

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- (v) But-1-ene is an isomer of but-2-ene.

Describe a chemical test that would enable you to distinguish between but-1-ene and but-2-ene.

Your answer should include the reagents and conditions for the test as well as relevant observations. Write balanced equations for any reaction that occurs.

[11]

[Total: 22]

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